

STATISTICS · SLIDE SET D0 · INTRO

Descriptive Statistics: Getting Started

What it is for — and the one distinction that shapes everything

This short guide unpacks the introductory deck to the Descriptive Statistics chapter. It answers three foundational questions before any formula appears: **what** is the goal of descriptive statistics, **which type** of variable are you looking at (and why that decides everything you do next), and **how** we summarise data — numerically *and* graphically. Get these straight and the rest of the chapter falls into place.

Summarising sample data

Descriptive vs inference

Qualitative vs quantitative

Discrete vs continuous

Numerical & graphical summaries

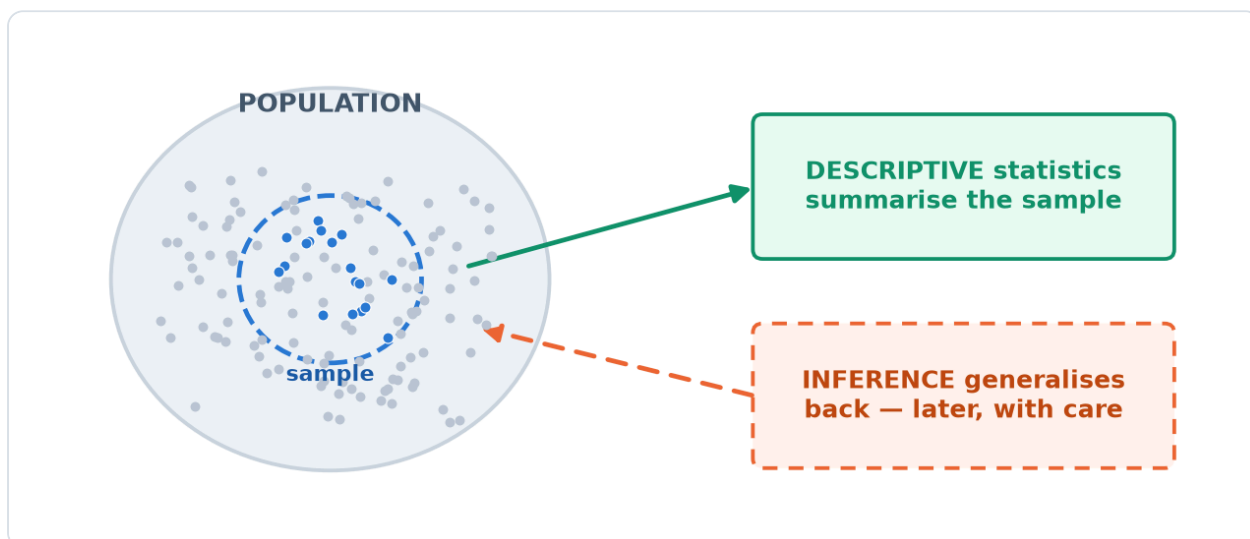
What descriptive statistics is for

Descriptive statistics has a single, modest job: **take a sample of data and summarise it** so a human can actually see what is going on. It does **not** — yet — try to prove anything about the wider world. Every technique in this chapter answers one of just **two questions**:

The two summary questions	What it means
1 · Distribution of outcomes?	Which values occur, and how often? (frequency tables and graphs — e.g. bar chart, needle graph, histogram)
2 · Summary measures?	Can we compress the data into a few numbers — a centre, a spread, ...?

THE GOLDEN RULE OF THIS CHAPTER

Do not immediately generalise your conclusions to the population. Descriptive statistics *describes the sample in front of you*. Extending those conclusions to the whole population is a **separate** job called **inference**, which comes later and needs extra tools (and caution). Confusing the two is the classic beginner's mistake.



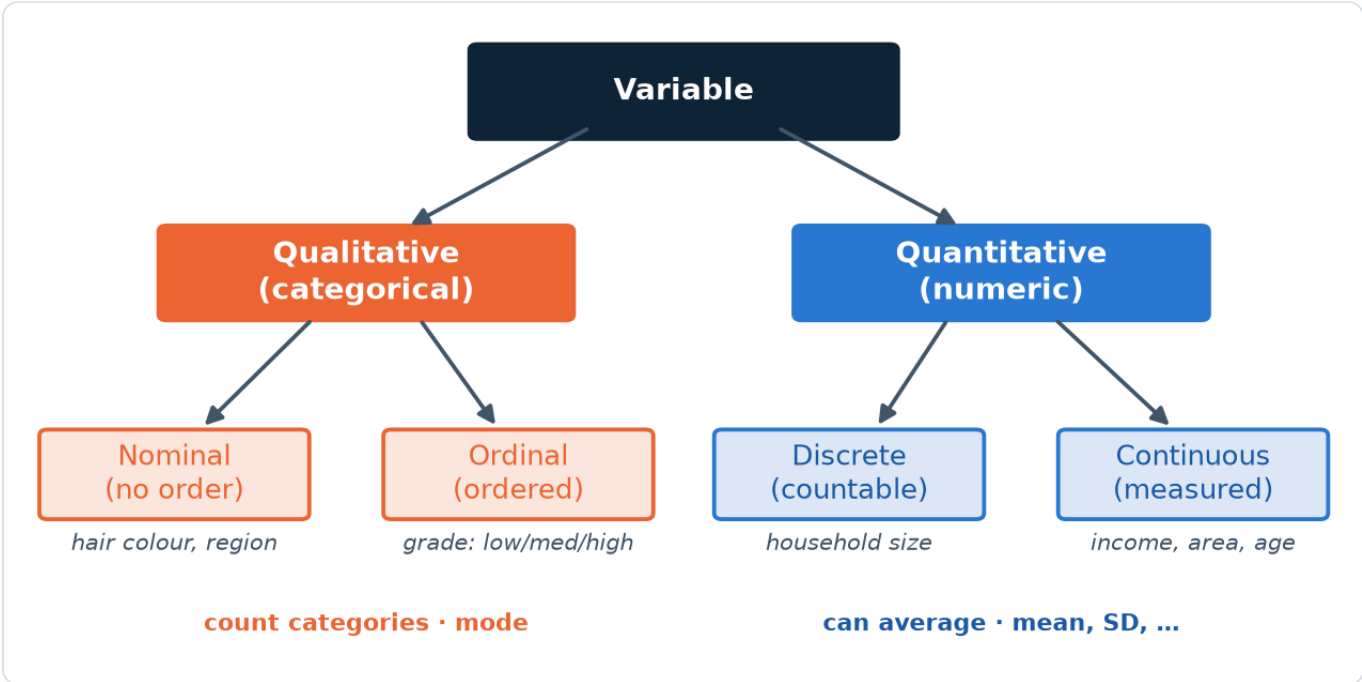
Descriptive statistics stays inside the sample (green). Generalising back to the population (orange) is inference — a later, more careful step.

HOW TO THINK ABOUT IT

If a question says “*summarise / describe / draw / compute the mean of these data,*” you are doing **descriptive** work — answer about **this sample**. If it says “*can we conclude that all X...*” or “*is this true in the population?*,” that is **inference** — a different chapter. Keep your answer scoped to what the data actually show.

The decision that shapes everything: variable type

Before choosing *any* graph or measure, ask one question: **what type is my variable?** The answer decides which tools even make sense. There are two families, and the quantitative one splits again:



The variable-type map. The split at the top drives every later choice of graph and summary measure.

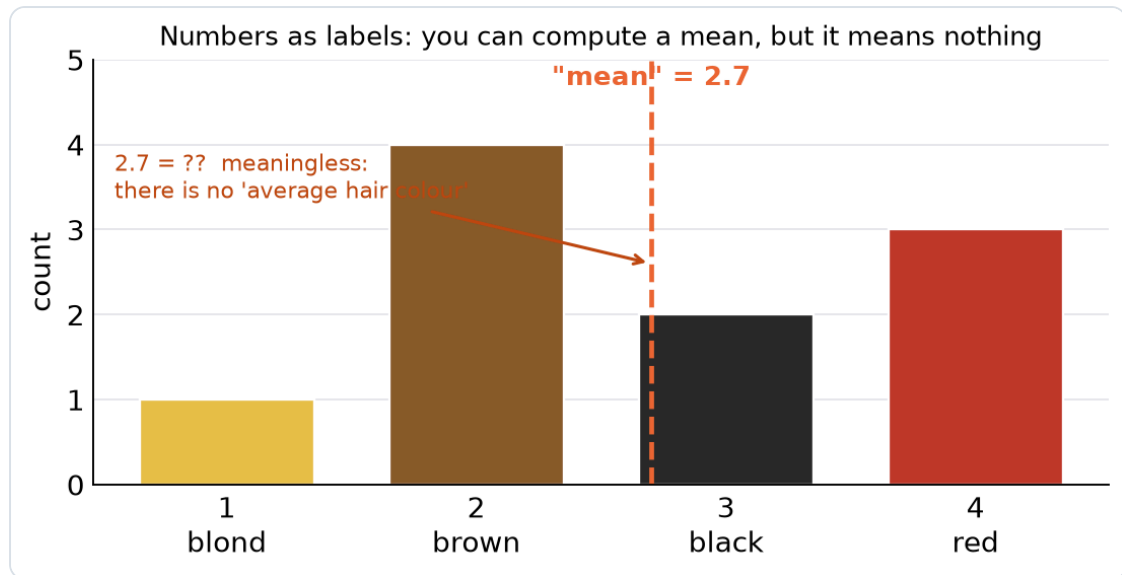
Type	What it is	Examples	What you may do
Qualitative (categorical)	Labels / categories. Nominal = no natural order; ordinal = ordered.	hair colour, region (nominal); grade low/med/high (ordinal)	count categories, proportions, mode ; bar charts
Quantitative (numeric)	Genuine numbers you can do arithmetic on. Discrete = countable; continuous = measured on a scale.	household size (discrete); income, area, age (continuous)	everything above plus mean, standard deviation, median, ...

WHY THE TECHNIQUE MUST MATCH THE TYPE

Some techniques *can* be applied to both types, but usually only **make sense for one**. You *can* feed category codes into an average formula — but the result is nonsense. Numbers used as **labels** are not numbers you can average.

THE HAIR-COLOUR EXAMPLE — WHY CODED CATEGORIES CAN'T BE AVERAGED

Hair colour is coded 1 = blond, 2 = brown, 3 = black, 4 = red. For 10 people you observe: **1, 4, 3, 2, 3, 2, 4, 2, 2, 4**. Nothing stops you computing the average of these numbers — $(1+4+3+2+3+2+4+2+2+4)/10 = 2.7$ — but 2.7 is **meaningless**: there is no “average hair colour,” and the codes could just as well have been letters. The only honest summary here is the **mode** (most frequent colour) or the counts/proportions per colour.



You can compute a mean of the codes, but it points at “2.7 = between black and... red?” — which is nonsense. Qualitative data → count and use the mode.

Two kinds of summary: numerical *and* graphical

Once you know the type, you summarise the data in **two complementary ways** — and good practice uses both.

Summary	What it is	Strength
Numerical	Compress the data to a few numbers — a proportion, a mean, a standard deviation, ...	precise, easy to compare and tabulate
Graphical	Show the data as a picture — bar chart, needle graph, histogram, boxplot, ...	reveals shape, outliers and patterns instantly

A PICTURE IS WORTH A THOUSAND WORDS

A single graph often shows skewness, gaps or outliers that no single number reveals — so **pair a numerical summary with a graph** rather than trusting either alone. (Later in the chapter, Anscombe’s quartet shows four datasets with *identical* numbers but completely different pictures — proof that you must always plot.)

Comparing several variables

When you summarise **more than one variable**, collect the numerical summaries into a **table** — one row per variable, columns for mean, standard deviation, etc. That keeps everything aligned and directly comparable at a glance.

HOW TO THINK IN THE EXAM

- **First move on any question:** classify the variable (qualitative / quantitative → nominal-ordinal / discrete-continuous). It rules in or out every tool.
- If a computed number looks odd (an “average” of category codes), ask whether the variable was really quantitative — you may be averaging **labels**.
- Give **both** a numerical summary and, where asked, the matching graph; justify your choice of measure by the variable’s type.
- Keep conclusions about the **sample** unless the question explicitly moves to the population (inference).

Where this sits in the chapter

This intro is the doorway to the rest of Descriptive Statistics:

Next up	Covers
D2 One quantitative variable	distribution (needle graph, histogram), mean/variance/SD, median & percentiles, boxplots, transformations, binned data
D3 Two variables together	contingency tables, side-by-side boxplots, scatterplots, covariance & correlation, linear combinations

Guide compiled from course slide set D0 — “Descriptive statistics of one variable: intro.” Diagrams are original illustrations of the deck’s ideas (variable taxonomy, sample-vs-population, and the hair-colour example).